

Lecture 6 Tutorial — Correlation and Regression

Q1. Definitions (Fill in the blanks)

A correlation exists between two variables when the values of one variable are somehow _____ with the values of the other variable.

A linear correlation exists when the paired data can be approximated by a _____ line.

Q2. Computing from Summary Statistics (Calculation)

Given

$$n = 5, \quad \sum x = 10, \quad \sum y = 20, \quad \sum x^2 = 30, \quad \sum y^2 = 90, \quad \sum xy = 47.$$

1. Compute r (to 2 decimal places).
2. State whether the correlation is positive, negative, or zero.

Q3. Properties of r (True / False)

Write T or F.

1. The value of r is always between -1 and 1 (inclusive).
2. If all x values are converted to a different scale (e.g., multiplied by a constant), the value of r changes.
3. Interchanging all x values and y values changes the value of r .

Q4. What r Measures (Multiple Choice)

Choose the single best answer.

Which statement is most accurate?

- A. r measures the strength of any relationship, including nonlinear relationships.
- B. r measures the strength of a linear relationship, and it is not designed to measure nonlinear strength.
- C. r equals the slope of the regression line.
- D. r is unaffected by outliers.

Q5. Sensitivity to Outliers (Multiple Choice)

Choose the single best answer.

A single outlier can dramatically affect:

- A. only the sample mean, not r
- B. only the sample standard deviation, not r
- C. the value of r
- D. the value of r^2

Q6. Critical r and Significance (Decision)

A study reports a computed correlation

$$r = 0.947.$$

The critical values are

$$r = \pm 0.666.$$

1. Is the correlation statistically significant by the critical-value rule? (Yes / No)
2. Briefly justify using an inequality (one line).

Q7. Spurious Correlation and Causality (Multiple Choice)

Choose the single best answer.

A high correlation value:

- A. proves x causes y
- B. proves y causes x
- C. does not by itself establish causality
- D. guarantees no confounding variables exist

Q8. Regression Coefficients from r (Calculation)

Given

$$r = 0.60, \quad s_x = 2, \quad s_y = 5, \quad \bar{x} = 10, \quad \bar{y} = 100.$$

Use:

$$b_1 = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

1. Compute b_1 .
2. Compute b_0 .
3. Write the regression equation in the form

$$\hat{y} = b_0 + b_1 x.$$

Q9. Prediction and Residual (Calculation)

Use the regression equation from Q8.

1. Compute the predicted value $\hat{y}$ when $x = 12$.
2. If the observed value is $y = 105$, compute the residual:

$$\text{Residual} = y - \hat{y}$$

3. State whether the point lies above or below the regression line.

Q10. Model Forms (Matching)

Match each regression type to its functional form.

Types:

- A. Linear
- B. Logarithmic
- C. Power
- D. Quadratic
- E. Exponential

Forms:

1. $y = a + bx$
2. $y = a + b \ln x$
3. $y = ax^b$
4. $y = ax^2 + bx + c$
5. $y = ab^x$

Write your answers as: A—____, B—____, C—____, D—____, E—____.

Q11. Logistic Regression (Exact Values, No Calculator)

Logistic regression uses:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

1. Compute $P(Y = 1)$ when $z = 0$.
2. Compute $P(Y = 1)$ when $z = \ln 3$. (*Hint: $e^{-\ln 3} = 1/3$*)

Using the decision rule “predict class 1 if $P(Y = 1) \geq 0.5$ ”, for which values of z is class 1 predicted?

- A. only $z > 0$
- B. $z \geq 0$
- C. only $z < 0$
- D. never (needs a different threshold)